AMS 572 Project

Benjamin Minkin, Marcus Minutello, Jacob White
AMS 572

Due: December 5, 2025 at 10:00am

What is our project about?

Bitcoin has evolved from a niche digital token into a globally traded financial asset:

- Held by hedge funds, Fortune 500 companies, and national banks.
- Increasing adoption as legal tender and payment medium (e.g., in El Salvador).
- Growth of broader cryptocurrency ecosystem.

Two cryptocurrency types:

- **Native coins** – technological improvements.
- **Memecoins** – speculative, hype-driven assets.

Questions of Interest

Question 1: Does Bitcoin's price movement follow that of the S&P 500?

Question 2: Do memecoins follow Bitcoin's price changes more than native coins?

Understanding these questions informs:

- Bitcoin's diversification value,
- The speculative nature of memecoins,
- Differences between crypto assets driven by "hype" and those driven by sought-after utility.

Data Description

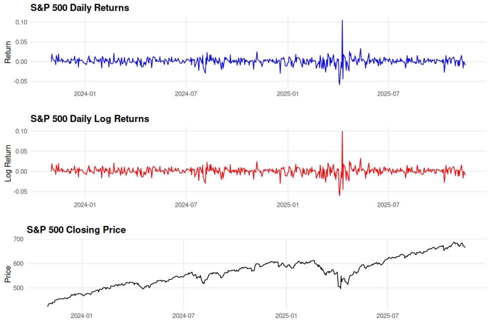
Data collected from Yahoo Finance using `quantmod` in R:

- Date range: Nov 1, 2023 – Nov 18, 2025.
- Log-returns computed via $r_t = \log(P_t / P_{t-1})$.
- Coins categorized as native vs. memecoin.

Background on Financial Time Series

A financial time series records prices or returns over time.

- Log-returns preferred for additivity and approximate normality.
- Efficient market hypothesis suggests log returns follow a random walk.



Correlation Between Time Series

Pearson correlation coefficient:

$$\hat{\rho} = \frac{\sum(x_t - \bar{x})(y_t - \bar{y})}{\sqrt{\sum(x_t - \bar{x})^2}\sqrt{\sum(y_t - \bar{y})^2}}$$

Used to test:

$$H_0 : \rho = 0.$$

BTC vs. Spy Log Returns

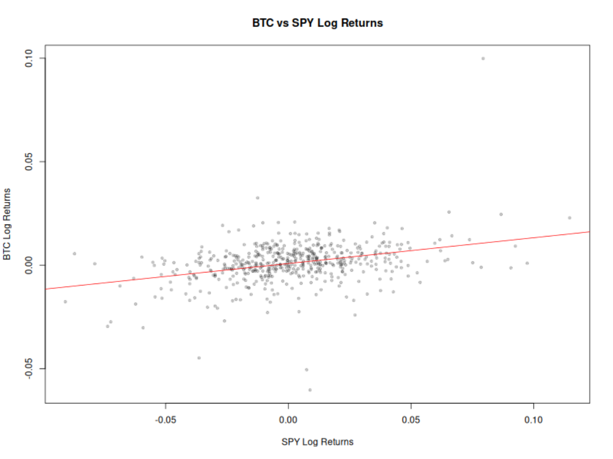


Figure 2: BTC vs. SPY log-return scatterplot.

Multiple Linear Regression with Interaction

Model:

$$\begin{aligned} \mathrm{Return}_{i,t} &= \beta_0 + \beta_1 \mathrm{BTC}_t + \beta_2 \mathrm{Memecoin}_i \\ &+ \beta_3 (\mathrm{BTC}_t \cdot \mathrm{Memecoin}_i) + \varepsilon_{i,t}. \end{aligned}$$

Interpretation:

- β_1: BTC sensitivity of native coins.
- β_3: **Additional** sensitivity of memecoins.

Multiple Linear Regression with Interaction

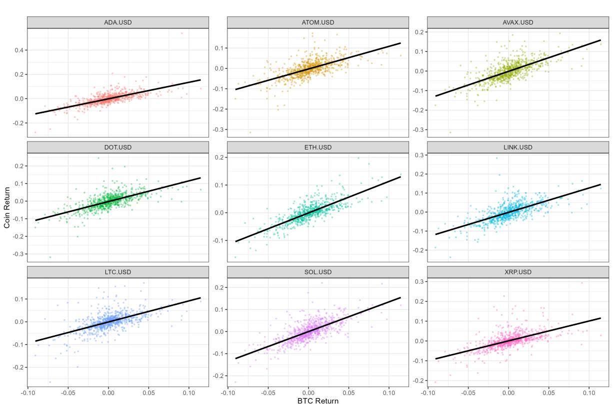


Figure 3: Native coin returns vs. BTC.

Memecoin Scatterplots

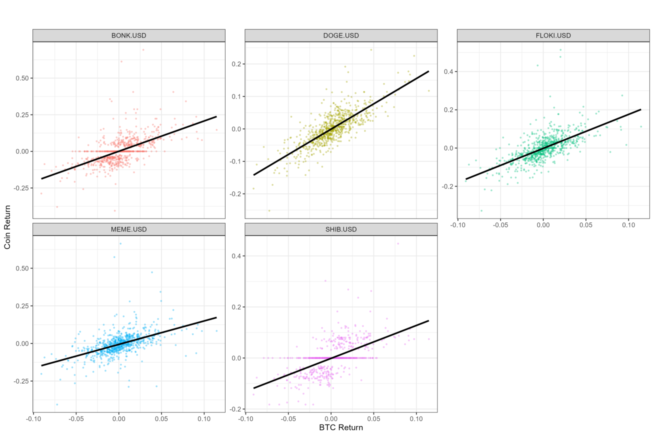


Figure 4: Memecoin returns vs. BTC.

Steeper slopes, but much greater volatility.

Distribution and Autocorrelation Diagnostics

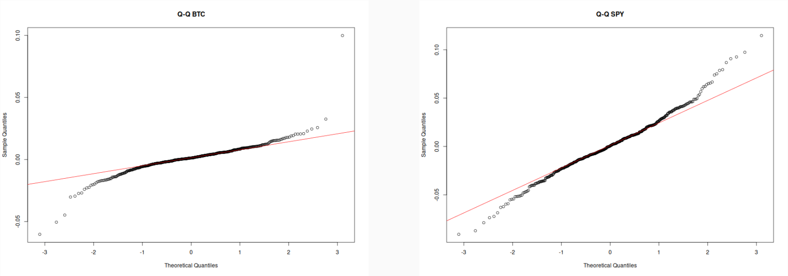


Figure 5: Q–Q plots for BTC and SPY.

ACF of BTC log-returns

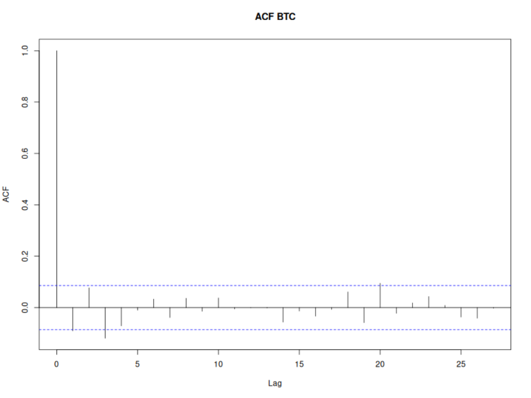


Figure 6: ACF of BTC log-returns.

Regression Diagnostics

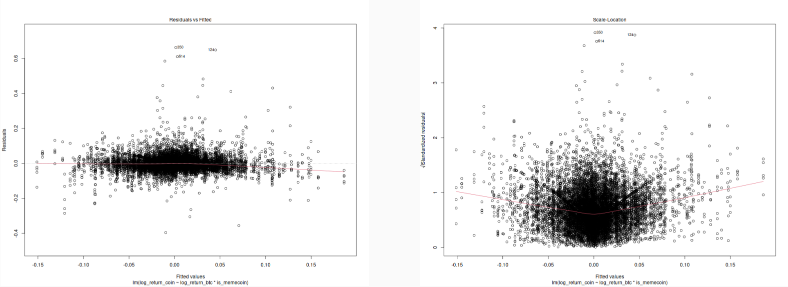


Figure 7: Left: Residuals vs. Fitted, **Right**: Scale-Location.

Question 1 Results

$$\hat{\rho} = 0.3542, \quad p = 4.79 \times 10^{-16}.$$

Conclusion: Bitcoin's returns significantly correlate with S&P 500 returns.

Missing data effects:

- MCAR: $\hat{\rho}_{MCAR} = 0.3807$
- MNAR: $\hat{\rho}_{MNAR} = 0.2795$

Correlation remained statistically significant in both cases.

Question 2 Results

$$\beta_3 = 0.4473, \quad p < 2 \times 10^{-16}.$$

Conclusion:

- Memecoins exhibit significantly stronger BTC sensitivity than native coins.

Missing data effects:

- MCAR: $\beta_{3,MCAR} = 0.4665$
- MNAR: $\beta_{3,MNAR} = 0.2213$

Summary of Findings

1. Bitcoin shows moderate correlation with the S&P 500.

2. Memecoins exhibit greater BTC sensitivity than native coins.

Implications:

- Bitcoin behaves increasingly like a traditional asset.
- Memecoins are much more sensitive to price changes in BTC than native coins.
- Native coins behave more like utility-driven blockchain assets.

Limitations & Future Work

Limitations:

- Violations of time-series assumptions.
- Two-year sample window.
- Selection bias.

Future work:

- Explore longer time horizons (this study was limited to about 2 years).
- Consider more crypto assets.
- Develop more detailed cryptocurrency categorization (i.e., consider properties of each cryptocurrency's implementation, and categorize based off these properties).

Thank you!